EXPERIMENT-5

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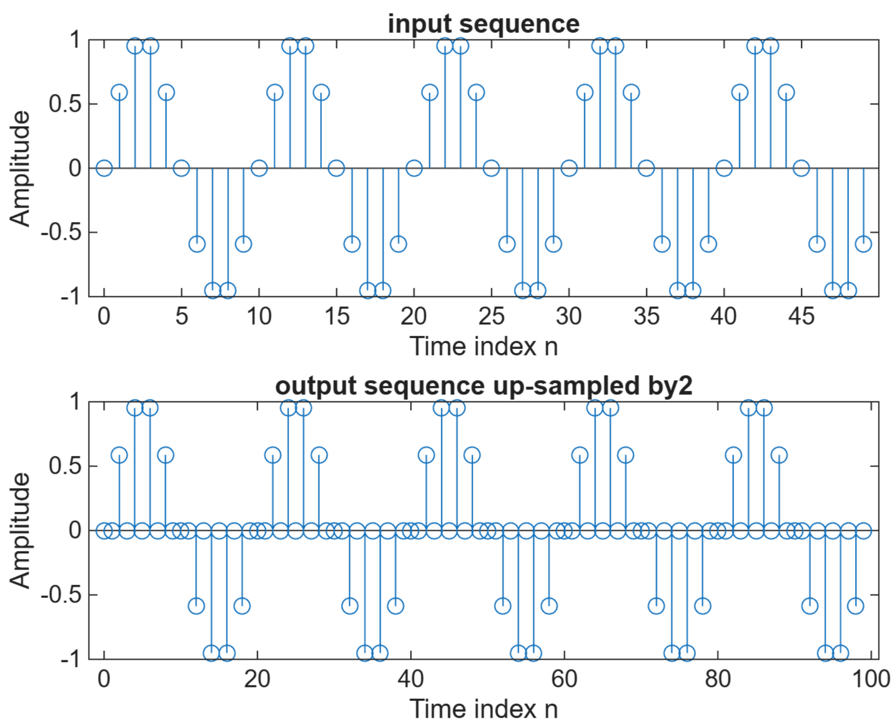
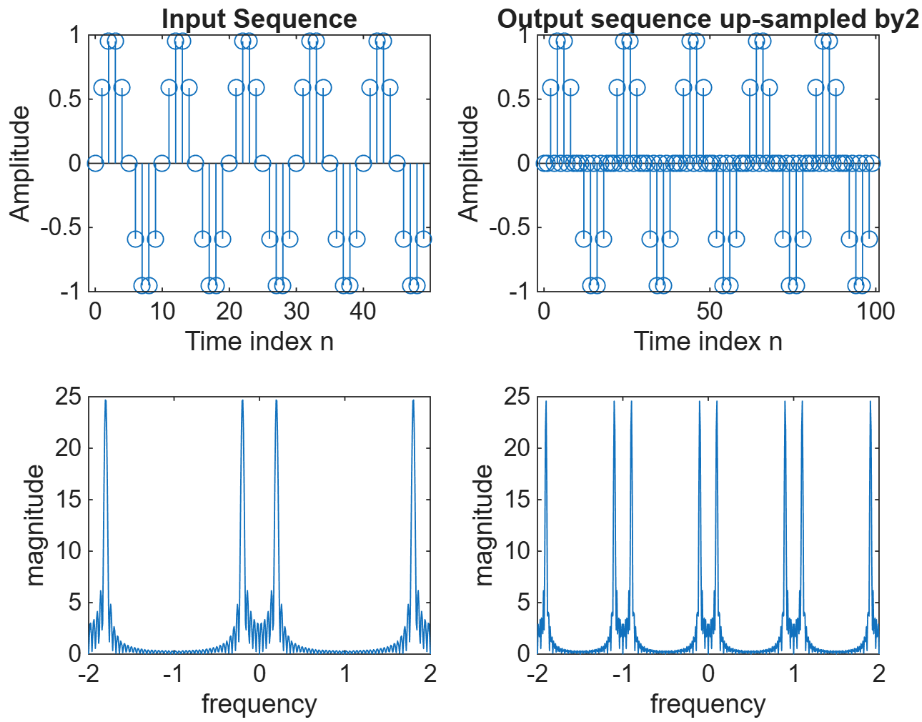
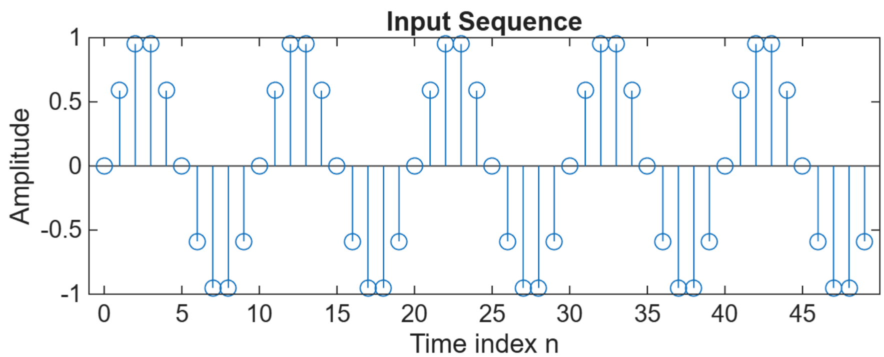
```
clc;clear all;close all;
N=input('Input length=');
L=input('Up-sampling factor=');
fo=input('Input signal frequency=');
n=0:N-1;
x=sin(2*pi*fo*n);
y=zeros(1,L*length(x));
y([1:L:length(y)])=x;
subplot(2,1,1)
stem(n,x);
xlabel('Time index n');
ylabel('Amplitude');
title("input sequence");
subplot(2,1,2)
stem(0:(L*N)-1,y(1:L*length(x)));
xlabel('Time index n');
ylabel('Amplitude');
title(['output sequence up-sampled by',num2str(L)]);
```

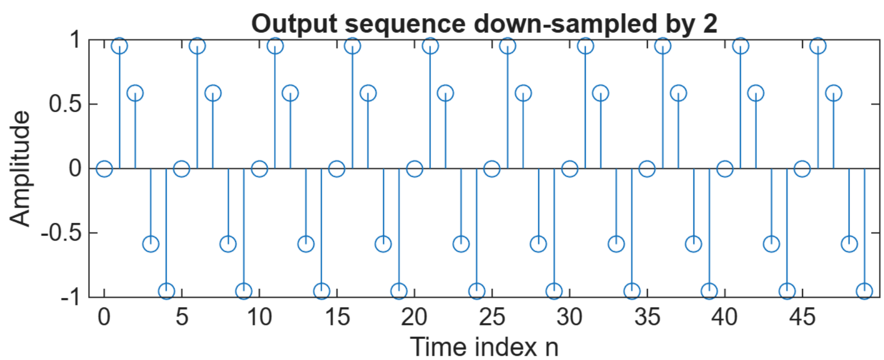


```
clc;
clear all;
close all;
N = input('Input length = ');
L = input('Up-sampling Factor = ');
fo = input('Input signal frequency = ');
n = 0:N-1;
x = sin(2*pi*fo*n);
y = zeros(1, L*length(x));
y([1: L: length(y)]) = x;
subplot(2,2,1);
stem(n,x);
xlabel('Time index n');
ylabel('Amplitude');
title('Input Sequence');
subplot(2,2,2);
stem(0:(L*N)-1,y(1:L*length(x)));
xlabel('Time index n');
ylabel('Amplitude');
title(['Output sequence up-sampled by', num2str(L)]);
f=-2:1/128:2;
h1=freqz(x,1,pi*f);
h2=freqz(y,1,pi*f);
subplot(2,2,3),
plot(f,abs(h1));
xlabel('frequency');
ylabel('magnitude');
subplot(2,2,4),
plot(f,abs(h2));
xlabel('frequency');
ylabel('magnitude');
```

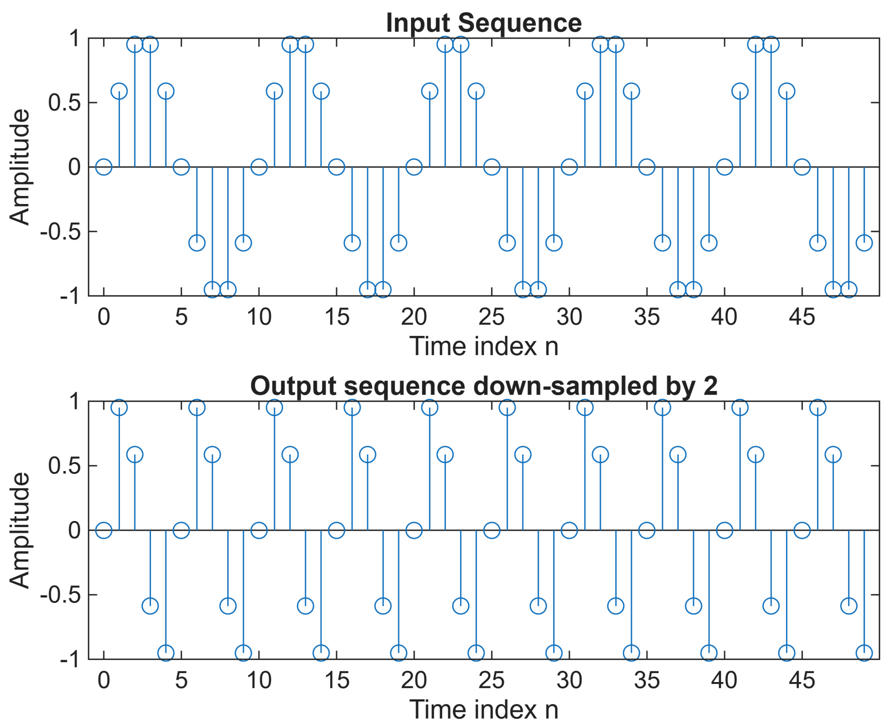

Input Sequence
Output sequence up-sampled by2
Amplitude
Time index n
magnitude
frequency

```
clc; clear all; close all;
N = input('Output length = ');
M = input('Down-sampling Factor = ');
fo = input('Input signal frequency = ');
n = 0 : N-1;
m = 0 : N*M-1;
x = sin(2*pi*fo*m);
y = x([1 : M : length(x)]);
subplot(2,1,1)
stem(n, x(1:N));
title('Input Sequence');
xlabel('Time index n');
ylabel('Amplitude');
subplot(2,1,2)
stem(n, y);
ylabel('Amplitude');
xlabel('Time index n');
title(['Output sequence down-sampled by ',num2str(M)]);
```





```
clc; clear all; close all;
N = input('Output length = ');
M = input('Down-sampling Factor = ');
fo = input('Input signal frequency = ');
n = 0 : N-1;
m = 0 : N*M-1;
x = sin(2*pi*fo*m);
y = x([1 : M : length(x)]);
subplot(2,1,1)
stem(n, x(1:N));
title('Input Sequence');
xlabel('Time index n');
ylabel('Amplitude');
subplot(2,1,2),
stem(n, y);
ylabel('Amplitude');
xlabel('Time index n');
title(['Output sequence down-sampled by ',num2str(M)]);
```




```
f=-2:1/128:2;
h1=freqz(x,1,pi*f);
h2=freqz(y,1,pi*f);
subplot(2,2,3),
plot(f,abs(h1));
xlabel('frequency');
ylabel('magnitude');
title('frequency domain representation of input sequence')
subplot(2,2,4),
```

```
plot(f,abs(h2));
xlabel('frequency');
ylabel('magnitude');
title('frequency domain representation of down-sampling sequence')
```